Practical No.: 3

Title: Applying Form Components in ReactJS

Aim: To demonstrate the use of form components in ReactJS, including input fields, textarea, select dropdown, and handling form submission.

Theory

Forms are an essential part of web applications. In ReactJS, forms work slightly differently than in traditional HTML. React controls form elements through **state** and updates them using event handlers.

Concepts Covered:

- Controlled Components
- Handling form input fields (text, textarea, select, radio buttons)
- Handling form submission

Syntax of Form Handling in ReactJS:

```
function ComponentName() {
  const [state, setState] = React.useState("");

  const handleChange = (event) => {
    setState(event.target.value);
  };
  return (
    <form>
      <input type="text" value={state} onChange={handleChange} />
    </form>
  );
}
```

.ReactJS Code

Step 1: Create `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Practical 3: Applying Form Components in ReactJS</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 3: Applying Form Components in ReactJS</h2>
    <div id="root"></div>

    <script type="text/babel" src="formValidation.js"></script>
</body>
</html>
```

Step 2: Create index.js

```
function SimpleForm() {
    const [name, setName] = React.useState("");
    const [comments, setComments] = React.useState("");
    const [gender, setGender] = React.useState("male");

    const handleSubmit = (event) => {
        event.preventDefault();
        alert(`Name: ${name} \nComments: ${comments} \nGender: ${gender}`);
    };

    return (
        <div>
            <h3>Fill the Form</h3>
            <form onSubmit={handleSubmit}>
                <label>
                    Name:
                    <input
                        type="text"
                        value={name}
                        onChange={(e) => setName(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
                    Comments:
                    <textarea
                        value={comments}
                        onChange={(e) => setComments(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
                    Gender:
                    <select value={gender} onChange={(e) =>
setGender(e.target.value)}>
                        <option value="male">Male</option>
                        <option value="female">Female</option>
                        <option value="other">Other</option>
                    </select>
                </label>
                <br /><br />

                <button type="submit">Submit</button>
            </form>
        </div>
    );
}
```

```
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<SimpleForm />);
```